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DEPARTMENT OF ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

EE 516 Power Systems

Project 2: Capability Curve and V-Curve Analysis of a Synchronous Generator

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1. Introduction

This project investigates the operating characteristics and limits of a synchronous generator using per-unit analysis and MATLAB visualization. The primary goal is to understand how real power, reactive power, field current, and armature current interact under different operating conditions.

A synchronous generator cannot operate arbitrarily at any point in the P - Q plane. Instead, its operation is constrained by physical and stability limits. These limits define the capability curve**,** which represents the safe region of operation.

In addition, the project explores **V-curves**, which describe how the armature current varies with field excitation for different levels of real power. These curves are essential for understanding generator behavior under under-excited and over-excited conditions.

2. Discussion

The results demonstrate the fundamental behavior of synchronous generators:

- Increasing field current increases reactive power output
- There exists a minimum excitation level below which the generator becomes unstable
- Armature current is minimized at unity power factor
- Both over-excited and under-excited operation increase current stress

The capability curve clearly illustrates the trade-offs between real and reactive power and highlights the physical limits of the machine.

Rated Operation

At rated operation, the generator is over-excited, meaning it supplies reactive power. In this case, the internal voltage is higher than the terminal voltage.

From the phasor relationship, the internal voltage and power angle are determined. The power angle shows how much the internal voltage leads the terminal voltage, and it directly affects power transfer.

The required field current at this point represents the normal operating condition of the generator.

Power Angle and Stability

When real power is kept constant, reducing the field current causes the power angle to increase. This happens because the machine must maintain the same power output.

As the field current continues to decrease, the power angle approaches 90^0. This is the stability limit**.** Beyond this point, the generator cannot stay synchronized and becomes unstable.

So, there is a minimum field current required to keep the generator stable.

Armature Current Behavior

Even when real power is constant, the armature current changes with field current. This is due to changes in reactive power.

At high excitation, the generator supplies reactive power, and the current is high. As excitation decreases, the current reduces and reaches a minimum near unity power factor.

If the field current is reduced further, the generator starts absorbing reactive power, and the current increases again.

Zero Real Power Case

When the generator produces no real power, the power angle becomes zero. In this condition, the current is purely reactive.

The armature current is minimum when the field current equals the rated value. On either side of this point, the current increases.

This produces a symmetric V-shaped curve.

V-Curve Set

When different real power levels are considered, multiple V-curves are obtained.

As the real power increases:

- The minimum current becomes larger
- The curves shift upward
- The generator operates over a smaller range of field current

These curves help show how the generator behaves under different loading conditions.

3) MATLAB Results

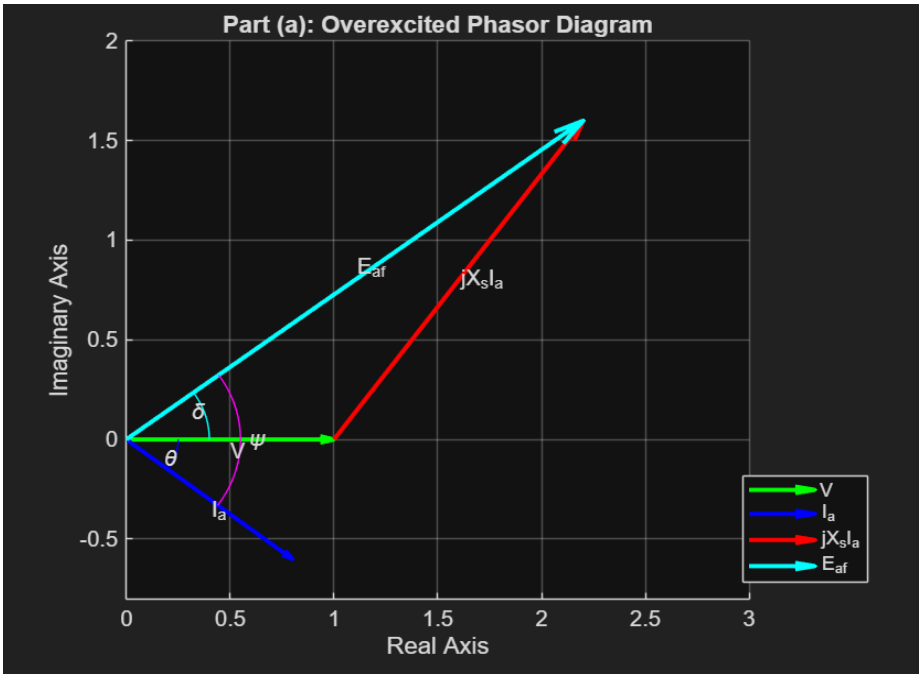

Figure 1: overexcited vector diagram of the synchronous generator

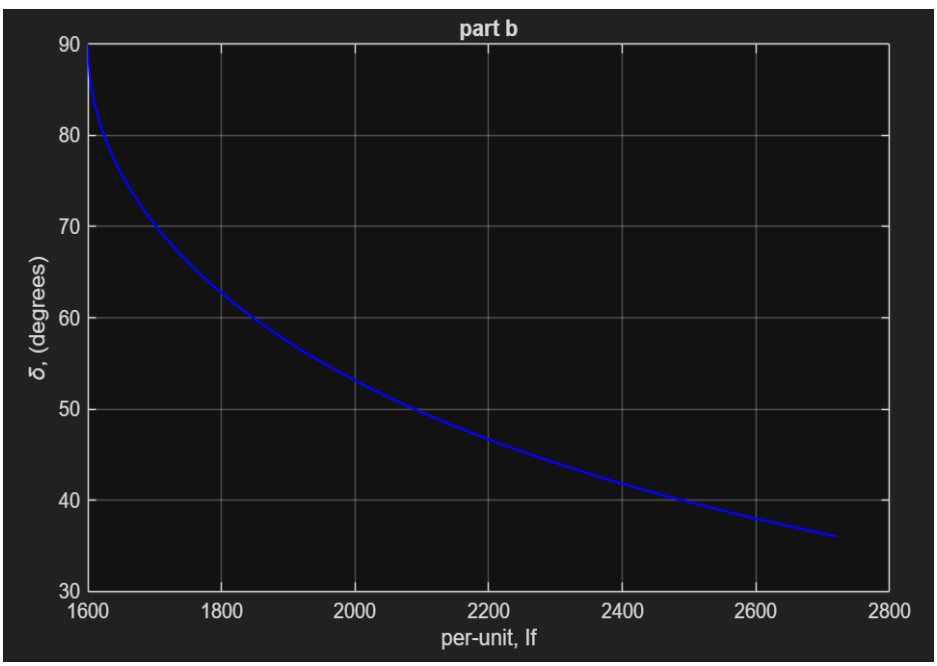

Figure 2:

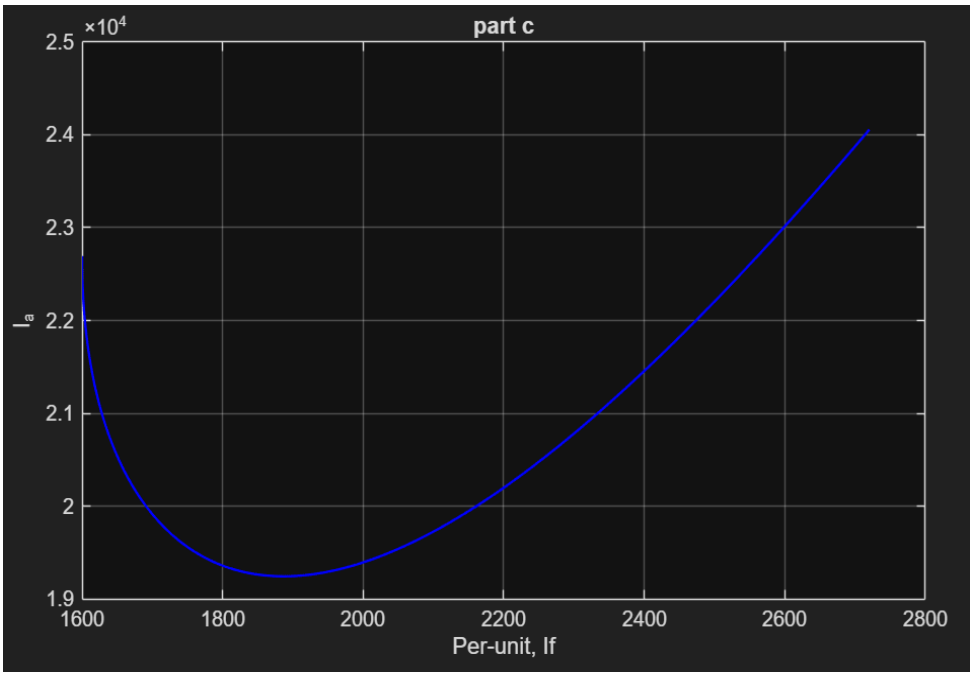

Figure 3

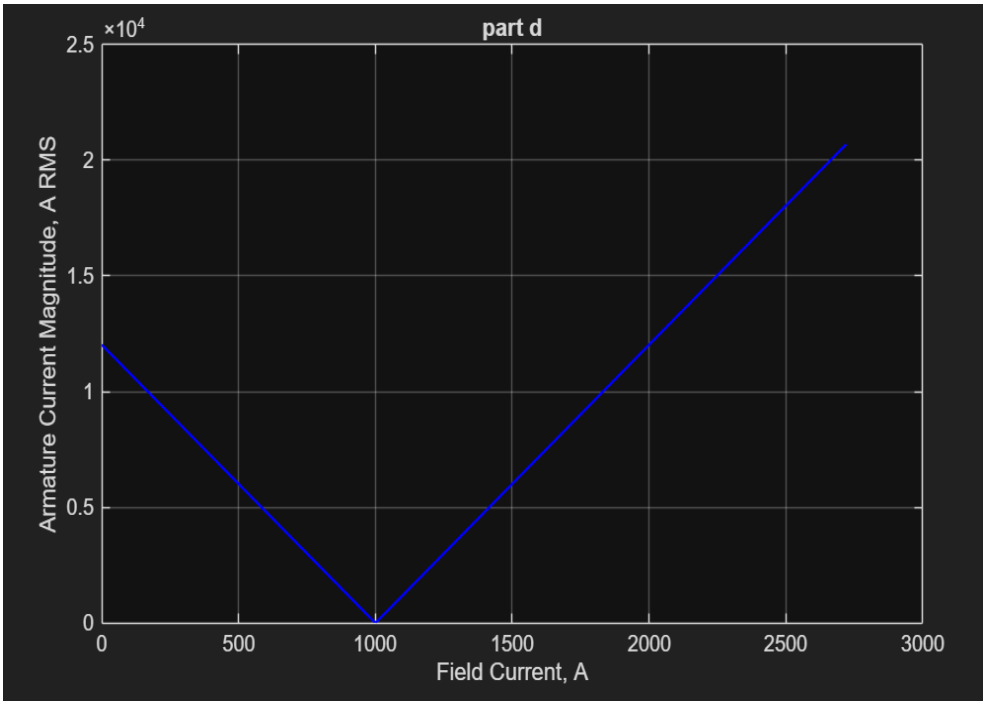

Figure 4

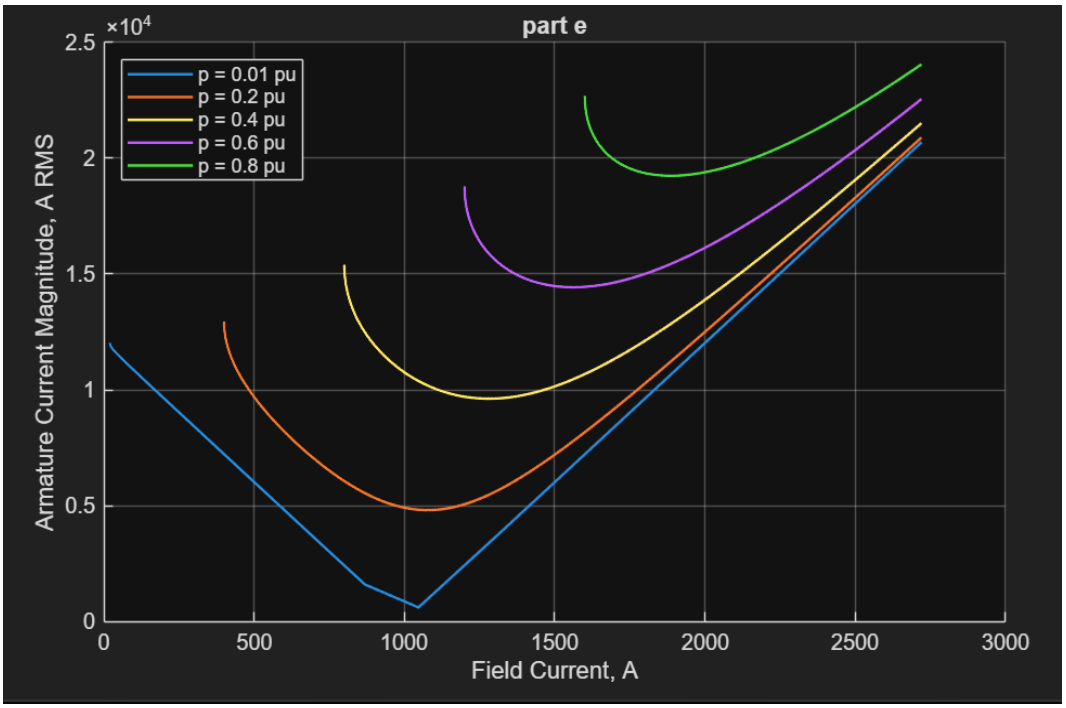

Figure 5

Question 2: Plot capability curve

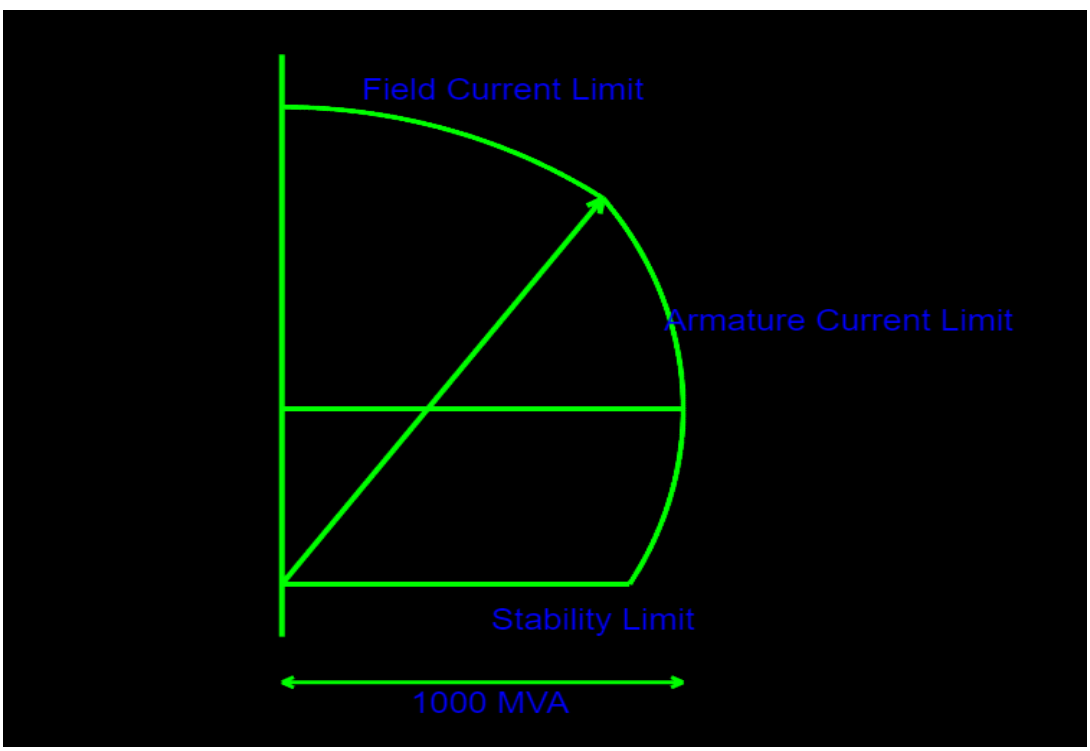

Figure 6: Capability Curve

4. Conclusion

This project helped me understand how a synchronous generator behaves under different conditions.

Some key observations:

- Field current controls reactive power
- There is a minimum excitation needed for stability
- Armature current is lowest at unity power factor
- Both over-excitation and under-excitation increase current

Overall, the results match the expected behavior of synchronous machines and clearly show their operating limits.

5) Appendix

These are the MATLAB codes

```
PowerAngleAnalysis.m
TorqueAngleVsFieldCurrent.m
RatedOperationVCurve.m
ZeroPowerVCurve.m
VCurveSetAnalysis.m
GeneratorCapabilityCurve.m
```